

EOX PFR

Example: The liquid phase reaction $2A \rightarrow B$ is being carried out isothermally in a PFR. The rate law is kC_A^2 and the rate constant equals $0.03 \text{ L/mol}\cdot\text{s}$. Species A is fed to the reactor at a concentration of 2 M and a flowrate of 3 L/s . Species A exits the reactor at a concentration of 0.1 M . Determine:

- The volume of the PFR.
- The rate of reaction in the PFR

mole balance: can use mole balance
in terms of F or X . Let's use
 F since information is given in terms
of F ; since doing so doesn't
increase the dimensionality of the
system.

$$\frac{dF_A}{dV} = r_A$$

rate law: $-r_A = kC_A^2$, so $r_A = -kC_A^2$

concentration equation: $C_A = \frac{F_A}{v}$

combining:

$$\frac{dF_A}{dV} = -k \frac{F_A^2}{v^2}$$

- use separation of variables method:
(constant (isothermal))

$$\frac{dF_A}{F_A^2} = -\frac{k}{v^2} dV$$

constant (liquid phase)

$$\int \frac{dF_A}{F_A^2} = -\frac{kV}{v^2}$$

- need limits of integration: $= 3/5$

$$C_A(V=0) = C_{A0} = 2 \text{ M} \quad F_{A0} = (2 \text{ M})v$$

$$F_{A0} = 6 \text{ mol/s}$$

$$C_A(V=V) = 0.1 \text{ M} \quad F_A(V=V) = 0.3 \frac{\text{mol}}{\text{s}}$$

- now we can integrate:

Solving in terms of Concentration

$$F_A = C_A v_0$$

$$\frac{dF_A}{dV} = v_0 \frac{dC_A}{dV} = r_A = -kC_A^2$$

$$v_0 \frac{dC_A}{dV} = -kC_A^2$$

$$\frac{dC_A}{C_A^2} = -\frac{k}{v_0} dV$$

$$-\frac{1}{C_A} \Big|_{C_{A0}}^{C_A} = -\frac{k}{v_0} dV$$

$$\int_6^{0.3} \frac{dF_A}{F_A^2} = -\frac{kV}{v^2}$$

$$\int \frac{dF_A}{F_A^2} = -\frac{1}{F_A}$$

$$-\frac{1}{F_A} \Big|_{0.3 \text{ mol/s}} + \frac{1}{F_A} \Big|_6 = -\frac{(0.02 \text{ mol/s})(V)}{(3 \text{ m}^3/\text{s})^2}$$

$$\left(\frac{1}{6 \frac{\text{mol}}{\text{s}}} - \frac{1}{0.3 \frac{\text{mol}}{\text{s}}} \right) \left(9 \frac{\text{L}^2}{\text{s}^2} \right) = V$$

$$\therefore V = 950 \text{ L}$$

reaction rate in the PFR depends on volume,

$-r_A = k C_A^2 = k \frac{F_A^2}{v^2}$, where

$$\left(\frac{1}{F_A} - \frac{1}{F_{A0}} \right) \left(\frac{v^2}{k} \right) = V, \text{ so}$$

$$\frac{V k}{v^2} + \frac{1}{F_{A0}} = \frac{1}{F_A} \quad \text{so} \quad F_A = \frac{1}{\frac{V k}{v^2} + \frac{1}{F_{A0}}}$$

$$-\left(\frac{1}{C_A} - \frac{1}{C_{A0}} \right) = -\frac{k}{v_0} V$$

$$\left(\frac{1}{C_A} - \frac{1}{C_{A0}} \right) \frac{v_0}{k} = V$$